DOG4DAG:
Semi-automated Ontology Generation in OBO-Edit and Protégé

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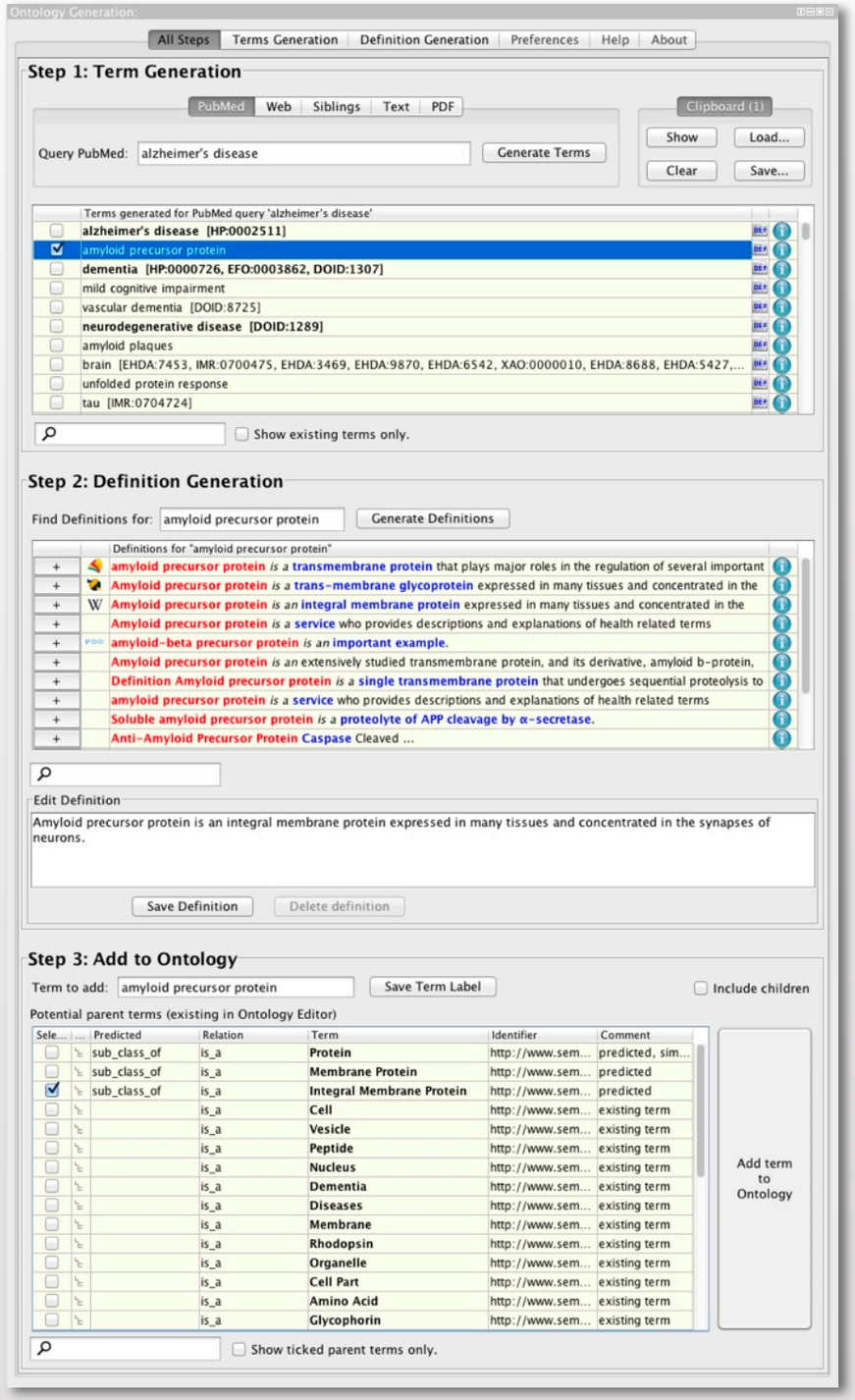
SWAT4LS 2011, London



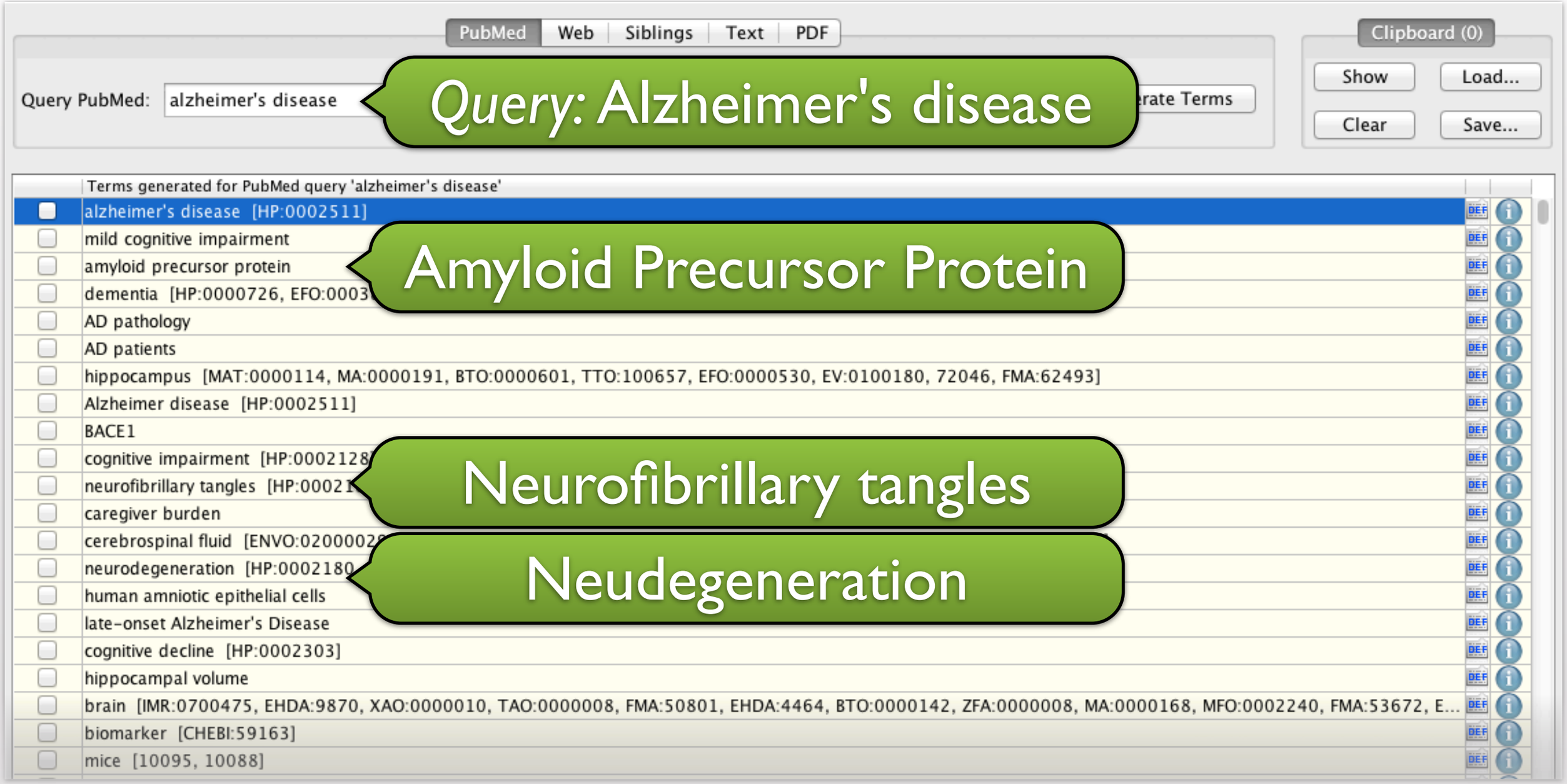


DOG4DAG

- Term generation
- Sibling generation
- Definition generation
- Relationship induction



Term Generation



Query: Alzheimer's disease

Amyloid Precursor Protein

Neurofibrillary tangles

Neudegeneration

Definition Generation

Step 2: Definition Generation

Find Definitions for: amyloid precursor protein

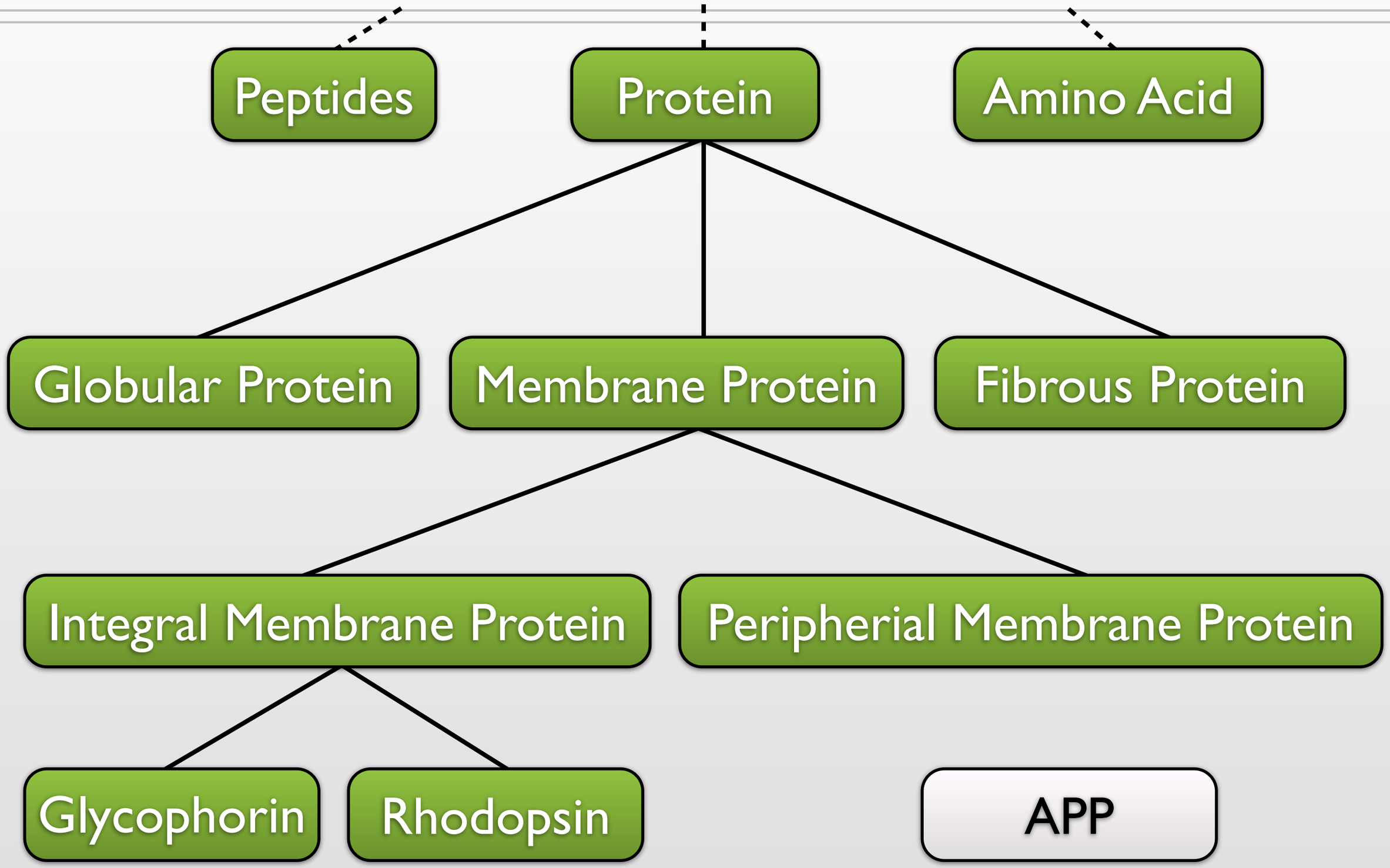
Query: Amyloid Precursor Protein

		Definitions for "amyloid precursor protein"	
+		**amyloid precursor protein** *is a* **transmembrane protein** that plays major roles in the regulation of several	
+		**Amyloid precursor protein** *is a* **trans-membrane glycoprotein** expressed in many tissues and concentrated in	
+	W	**Amyloid precursor protein** *is an* **integral membrane protein** expressed in many tissues and concentrated in the	
+		**Amyloid precursor protein** *is a* **service** who provides descriptions and explanations of health related terms	
+	PDB	**amyloid-beta precursor protein** *is an* **important example.**	
+		**Amyloid precursor protein** *is an* extensively studied transmembrane protein, and its derivative, amyloid	
+		**Definition Amyloid precursor protein** *is a* **single transmembrane protein** that undergoes sequential proteolysis	
+		**Soluble amyloid precursor protein** *is a* **proteolyte of APP cleavage by α-secretase.**	
+		Activation of caspases and abnormal processing of **amyloid precursor protein** *are* **important events in other chronic**	
+		**Anti-Amyloid Precursor Protein Caspase** Cleaved ...	

Amyloid precursor protein is an **integral membrane protein** expressed in many tissues and concentrated in the synapses of neurons.

[http://en.wikipedia.org/wiki/Amyloid_precursor_protein]

Relationship Induction



Relationship Induction

Term to add: amyloid precursor protein Save Term Label

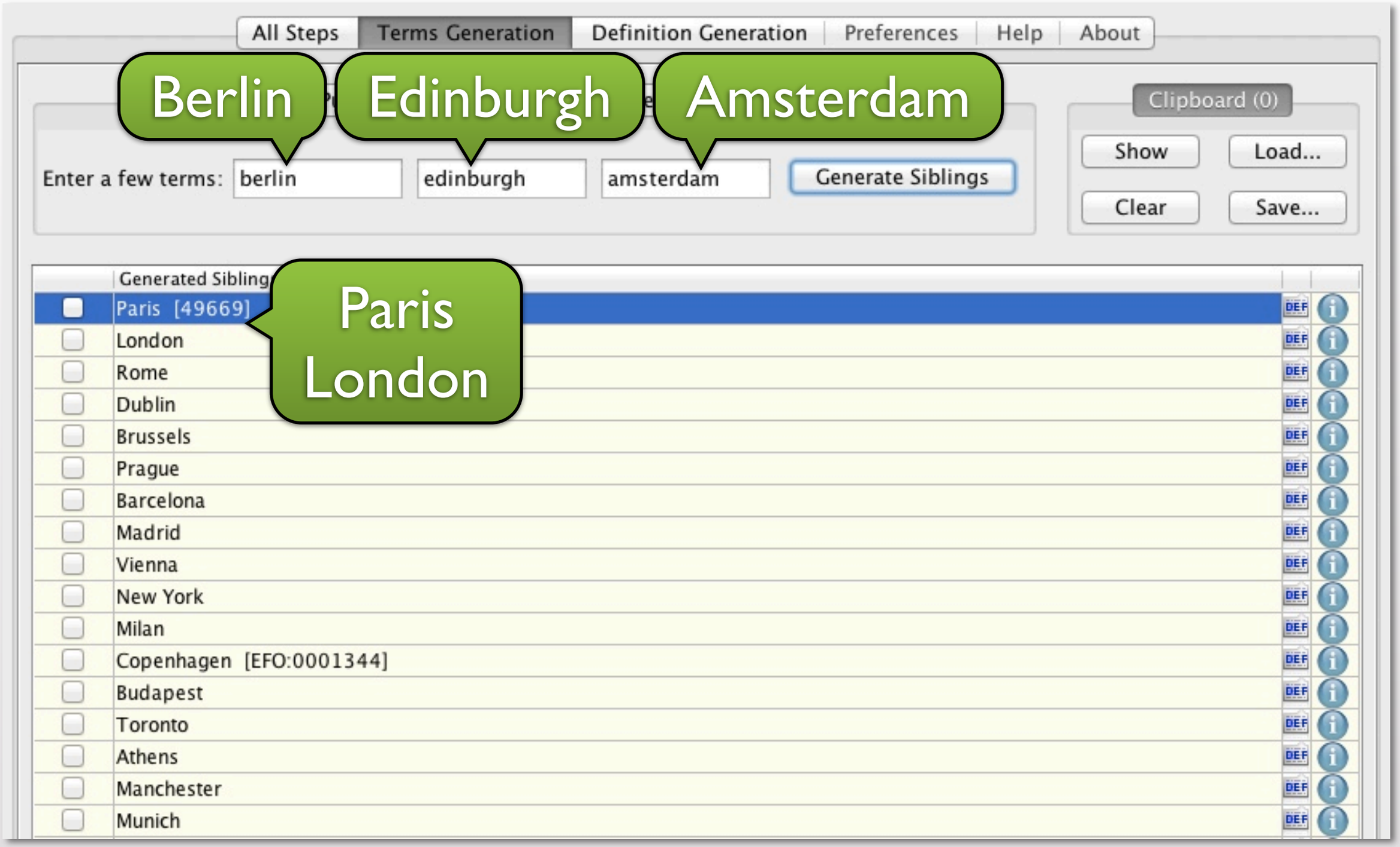
Potential parent terms (existing in Ontology Editor)

Sele...	...	Predicted	Relation	Term	Identifier	Comment
☐	⊢	sub_class_of	is_a	**Protein**	http://www.se...	predicted, sim...
☐	⊢	sub_class_of	is_a	**Membrane Protein**	http://www.se...	predicted
☑	⊢	sub_class_of	is_a	**Integral Membrane Protein**	http://www.se...	predicted
☐	⊢		is_a	**Peptide**	http://www.se...	existing term
☐	⊢		is_a	**Dementia**	http://www.se...	existing term
☐	⊢		is_a	**Diseases**	http://www.se...	existing term
☐	⊢		is_a	**Rhodopsin**	http://www.se...	existing term
☐	⊢		is_a	**Amino Acid**	http://www.se...	existing term
☐	⊢		is_a	**Glycophorin**	http://www.se...	existing term
☐	⊢					m
☐	⊢					m
☐	⊢					m
☐	⊢					m
☐	⊢					m
☐	⊢					m

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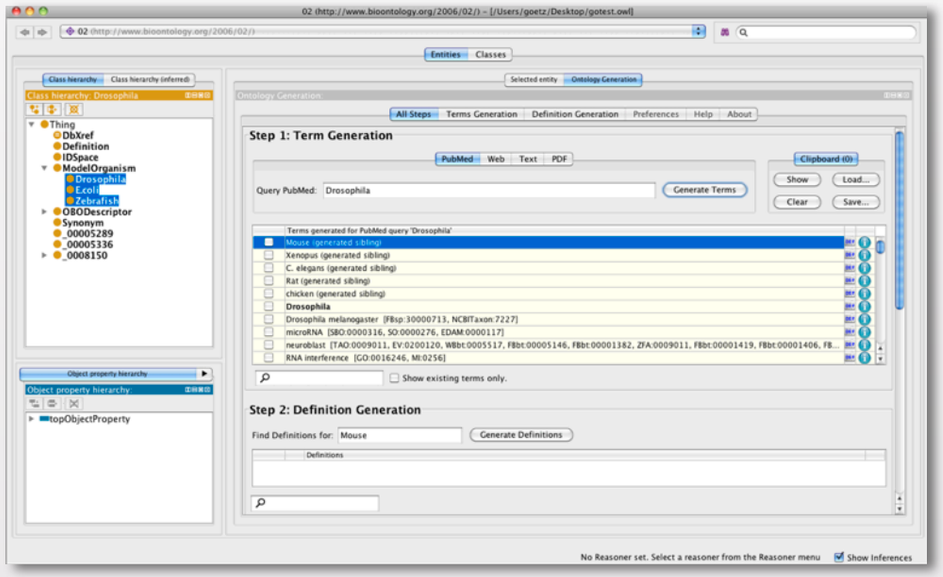
[http://en.wikipedia.org/wiki/Amyloid_precursor_protein]

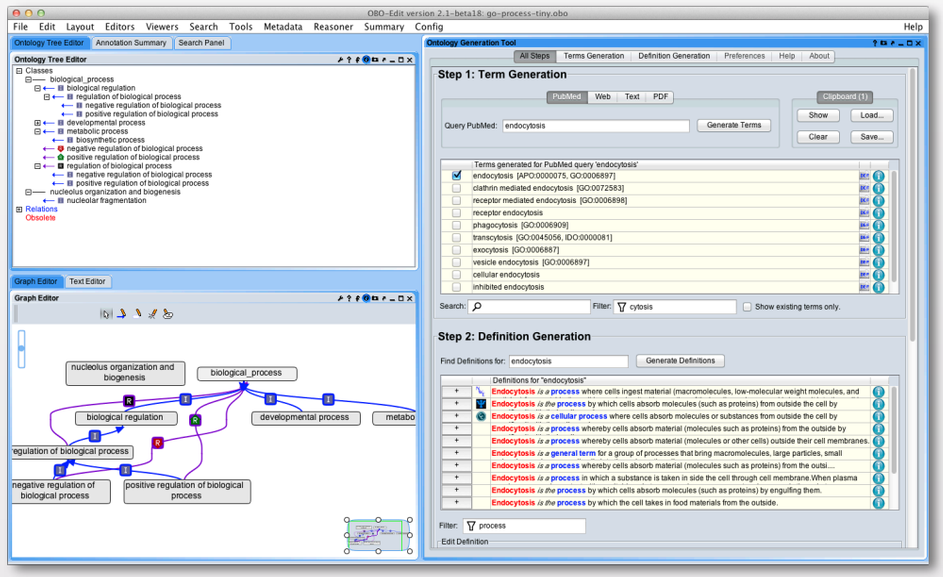
DOG4DAG says: SWAT4LS should be in Paris!



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Works in both Protégé and OBO-Edit





Find me at the poster session!

Read More: Wächter *et al.*, Bioinformatics 2010

Google *DOG4DAG Dresden*

Contact

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